Assume/Guarantee Contract of `CalculateGyroDg`

May 22, 2026

1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet : JoinTotal, SignFlag, wa, Rtemp, W′, clock, clock′

Inputs :

- `JoinTotal` $\mapsto$ JoinTotal : $\mathbb{N}$
- `SignFlag` $\mapsto$ SignFlag : $\mathbb{N} \rightharpoonup \mathbb{N}$
- `wa` $\mapsto$ wa : $\mathbb{N} \rightharpoonup \mathbb{R}$
- `Rtemp` $\mapsto$ Rtemp : $(\mathbb{N}, \mathbb{N}) \rightharpoonup \mathbb{R}$

Outputs :

- `W` $\mapsto$ W : $\mathbb{N} \rightharpoonup \mathbb{R}$

In-Outputs : None

States : None

Parameters :None

Clocks : clock, clock′ : $\mathbb{R}_{\geq 0}$

2 AG Contract

Assumptions :

$$
\begin{aligned}
Assumptions \triangleq\ & \mathsf{len}(\mathsf{SignFlag}) \geq \mathsf{JoinTotal} \\
& \land\ \mathsf{JoinTotal} \leq 5 \\
& \land\ \forall 0 \leq j < \mathsf{JoinTotal} \cdot \mathsf{SignFlag}[j] < \mathsf{len}(\mathsf{wa}) \\
& \land\ \mathsf{size}(\mathsf{Rtemp}) = (3, 5)
\end{aligned}
$$

Guarantees :

$$
\begin{aligned}
Guarantees \triangleq\ & Check_1 \land Check_2 \\
Check_1 \triangleq\ & \mathsf{len}(\mathsf{W}') = 3 \land \mathsf{clock}' = \mathsf{clock} \\
Check_2 \triangleq\ & \exists \mathsf{tmpwa} : \mathbb{N} \rightharpoonup \mathbb{R} \cdot \big(\ \mathsf{len}(\mathsf{tmpwa}) = 5\ \land\ Check_3 \land Check_4 \land Check_5\ \big) \\
Check_3 \triangleq\ & \forall 0 \leq j < \mathsf{JoinTotal} \cdot \mathsf{tmpwa}[j] = \mathsf{wa}[\mathsf{SignFlag}[j]] \\
Check_4 \triangleq\ & \forall j \geq \mathsf{JoinTotal} \cdot \mathsf{tmpwa}[j] = 0 \\
Check_5 \triangleq\ & \mathsf{if}\ \mathsf{JoinTotal} \geq 3\ \mathsf{then} \\
& \quad \mathsf{W}' = \mathsf{Rtemp} \bullet \mathsf{tmpwa} \\
& \mathsf{else} \\
& \quad \forall 0 \leq i < 3 \cdot \mathsf{W}'[i] = 0 \\
& \mathsf{endif}
\end{aligned}
$$